

Assignment 1: E-Purchase Business Process Modeling

CSE346/CSE345 Business Process Modeling

Spring 2026

1 Scope and Tool Choice

The modeled process is an Amazon/Souq-like e-purchase order-to-cash flow. It starts when a customer submits checkout details and ends when the order is delivered, archived, or rejected. The model follows the BPM lifecycle from the lectures: define objectives, model the as-is process, analyze issues, design the to-be process, then prepare implementation and monitoring.

Primary online modeling tool: bpmn.io because Lecture 2 explicitly lists it as a BPMN tool and it supports BPMN 2.0 import/export in the browser. The generated BPMN XML files are:

- output/assignment1_epurchase_as_is.bpmn
- output/assignment1_epurchase_to_be.bpmn

Optional online validation/export tools: Camunda Web Modeler for BPMN editing and diagrams.net for final image layout if a single-page polished figure is required. The report keeps BPMN 2.0 semantics as the source of truth.

2 Lecture Concepts Applied

The report uses these concepts from the LECS folder:

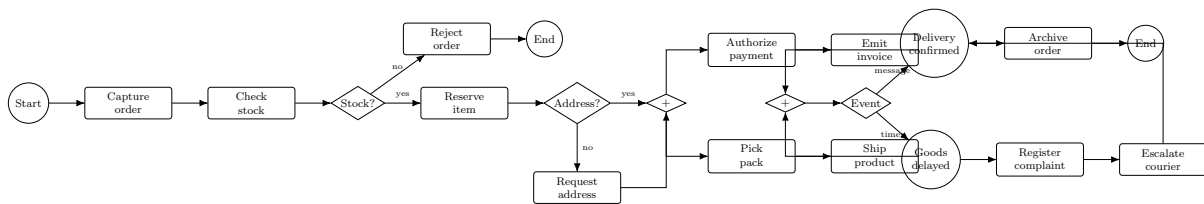
- BPM models make invisible work visible and support analysis before automation.
- BPMN Business Process Diagrams consist of flow objects, connections, swimlanes, and artifacts.
- XOR gateways model exclusive decisions; AND gateways model parallel work and synchronization.
- Pools and lanes represent participants, roles, departments, systems, and handoffs.
- Intermediate timer/message events represent events that occur during process execution.
- Process analysis evaluates cycle time, capacity, cost, quality, risk, and customer interfaces.
- Redesign patterns include eliminating waste, introducing parallelism, improving communication, using IT-driven improvements, and appointing process/case managers.

3 As-Is BPMN Model

3.1 Participants and Data

Participant or lane	Responsibility
Customer	Places order, provides address, receives invoice and delivery, may complain about delay.
Marketplace system	Captures order, checks stock, validates address, emits invoice, archives order.
Payment service	Authorizes payment and confirms payment status.
Warehouse	Reserves item, picks, packs, and prepares shipment.
Courier	Ships product and reports delivery status.
Customer support	Registers complaints and escalates delayed shipments.

3.2 As-Is Control Flow



The goods-delay requirement is modeled by an intermediate timer event after shipment. If delivery confirmation is not received within the SLA, the timer branch opens a complaint and escalates the courier. This follows the lecture definition of an intermediate timer event: it is enabled during the process and fires after a time interval elapses.

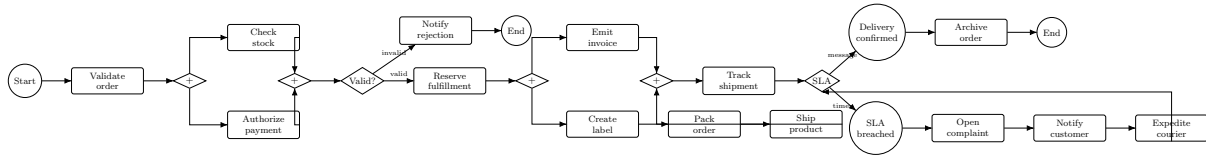
4 As-Is Analysis

No.	Issue	Consequence	Type	Proposed fix
1	Late complaint is reactive	Customer discovers delay before the company does; poor external quality.	Process / customer interface	Add proactive shipment tracking and automatic SLA monitoring.
2	Address collection may happen after stock reservation	Reserved stock can wait while customer data is corrected.	Data / process	Validate address at checkout before reservation.
3	Payment and fulfillment handoffs are loosely synchronized	Manual checks can delay either invoice or shipment.	IT / organization	Run payment authorization and stock checks in parallel.
4	Complaint path is separate from normal case ownership	Support may not have full order context.	Organization	Assign a case owner and attach complaint to the same order case.
5	Archive occurs only at the end	Operational data may be unavailable for monitoring until closure.	IT	Record status events throughout the process.

The main performance risks are waiting time and handoff delay. The lectures define cycle time as the time from process start to end; in this process, cycle time is affected by address

correction, payment confirmation, shipment waiting time, and complaint escalation. Several steps are business-value adding, such as payment control and archiving, but waiting for missing data and discovering delays late are waste.

5 To-Be BPMN Model



6 Enhancements and Comparative Analysis

Area	As-is	To-be improvement
Checkout data	Address may be requested after stock reservation.	Validate address, stock, and payment early.
Cycle time	Sequential checking increases waiting.	Stock and payment checks run in parallel, following the lecture redesign pattern of introducing parallelism.
Delay handling	Complaint starts only after a late delivery event.	SLA timer creates a proactive complaint and customer notification.
Quality	Customer may have to ask for status.	Tracking status is recorded and communicated automatically.
Cost and rework	Support may manually reconstruct order context.	Complaint is attached to the same order case with a case owner.
Flexibility	Same path for all orders after basic stock check.	Exceptions are explicit: invalid order, SLA breach, courier escalation, and final delivery.

Expected outcomes are shorter average cycle time, fewer avoidable waiting periods, better customer-facing quality, and improved monitoring. There is a trade-off from the process analysis lecture: the to-be model has more IT coordination and status events, but it reduces waiting, communication errors, and complaint handling cost.

7 Case Coverage

- Normal successful order: checkout, stock/payment validation, fulfillment, delivery, archive.
- Out-of-stock or invalid payment: notify rejection and end rejected.
- Missing or invalid address: caught at validation before stock is reserved.
- Shipment delivered on time: delivery message event continues to archive.

- Goods delayed: intermediate timer event opens complaint, notifies customer, escalates courier, and loops back to wait for delivery status.
- Operational audit: archive order and status events support monitoring and future process mining.

8 Submission Checklist

The zipped submission should include this report PDF, the two BPMN XML files, and exported images from bpmn.io or Camunda if required by the instructor. The BPMN files were generated by `tools/generate_assignment1_bpmn.py`, then can be imported into the selected online BPMN tool.